C++ Programming Basics

1] Algorithm to Add 2 no.s

Step I :- Take 2 no. a and b

Step II :- Ask user the value of a and b

Step III :- assign variable c = a + b

Step IV :- Print c

2] Greater variable will print.

Step I :- take 2 no. a and b

II :- Ask user the value of a and b

III :- if a is greater than b, print a

IV :- if b is greater than a ; Print b

V :- if a = b , print "Both variables are equal"

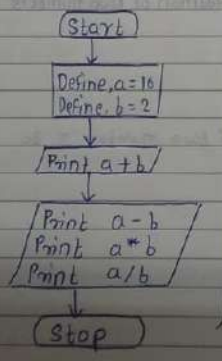
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Cw:- 5]

```
#include <iostream>
using namespace std;
int main()
{
  int a = 10;
  int b = 2;
  Cout << "Addition of two no.s =" << a+b << endL;
  Cout << "Subtraction =" << a-b << endL;
  Cout << "multiplication =" << a*b << endL;
  Cout << " Division =" << a/b;
  return 0;
}
```

o/p :- Addition of two no.s = 12
Subtraction = 8
multiplication = 20
& division = 5

Hw :- Flow chart :-



3 o/p :-
12
8
20
5

" Flow chart :-

⟶ :- Flow line :- indicate flow of logic by connecting symbols

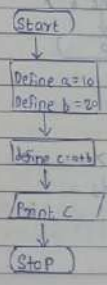
(Start) :- Start / stop :- indicate starting / ending points

▭ :- Processing :- indicates data manipulation

▱ :- I/O :- used for input / output operations

◇ :- Descision :- used for decision making between two or more alternatives

1] Adding 2 variables :-

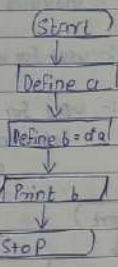


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Q.no 6] Square of a number

Algorithm :- ① Take one ~~number~~ variable a
② Ask user value of a
③ Multiply a by itself, assign variable b=a*a
④ Print b

Flowchart :-



Program :-

```
#include <iostream>
using namespace std;
int main()
{
 int a = 10
 int b = a*a
 Cout << "Square of "<<a<<"="<<b;
 return 0;
}
```

Q.no 7] :- o/p :- Roll no = 4
** Subject Marks **
Maths = 75
Phy = 72
Chem = 82
Avg =
-Percentage = -

```
# include <iostream>
using namespace std;
int main ()
{
  int a = 75
  int b = 72
  int c = 82
 Cout << " Roll no = 4 " <<endL ;
 Cout << " ** Subject Marks ** " <<endL ;
 Cout << " Maths = " << a <<endL ;
 Cout << " Phy = " <<b <<endL ;
 Cout << "Chem = "<< c <<endL ;
 Cout << "Avg = " << (a+b+c)/3 <<endL ;
 Cout << "Percentage = " << (229/300)* 100 << endL ;
  return 0 ;
}
```

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Program 1:-

```
# include <iostream>
using namespace std;
int main()
{
    cout << "Hello students";

    return 0;
}
```

- used for I/P and O/P
- skips std:: when using cout / cin
- int main → starting point
- { } → Begining of program
- cout → display O/P
- return 0 → Successful execution.

C.Q.:- 2]

```
# include <iostream>
using namespace std;
int main()
{
  cout << "Anushree Pansare";
  cout << " FYIT";
  cout << "Roll no 21";
  return 0;
}
```

Output :-

Anushree Pansare FYIT
Roll no 21

```
# include <iostream>
using namespace std;
int main()
{
  cout << "Anushree Pansare" << endl;
  cout << " FYIT " << endl;
  cout << "Roll no 21" << endl;
  return 0
}
```

output :-

Anushree Pansare
FYIT
Roll no 21

3]

```
# include<iostream>
using namespace std;
int main()
{
  cout "Addition of two numbers = " <<10+20;
  return 0;
}
```

O/P :- Addition of two numbers = 30

4]

```
# include <iostream>
using namespace std;
int main()
{
  int a = 10;
  int b = 20;
  int c = a+b;
  cout << "Addition of two numbers = ";
  cout c;
  return 0;
}
```

O/P :- Addition of two number = 30

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- c++ :- general purpose, high-level, programming language built as extension of C by adding OOP features.

- History :-

1972 1979 1983 1985 1998 2011

note :- further part will be in H.W / C.W Part.

C.W :- Grade teller :

Step I :- Take / ask user for obtained marks.

Step II :- Assign grade as per the marks obtained depending on the condition :

if marks $\geq$ 75 assigned grade is A+
if marks is between 60 to 75 assign grade A
if 50 $\leq$ marks $\leq$ 60 assigned grade is B
if 40 $\leq$ marks $<$ 50 assigne. grade C
if marks below 40 assign grade F

Step III :- Return assigned grade as per the condition.

a = 10 ↓ assigned a == 10 ↓ compare/relation operator.

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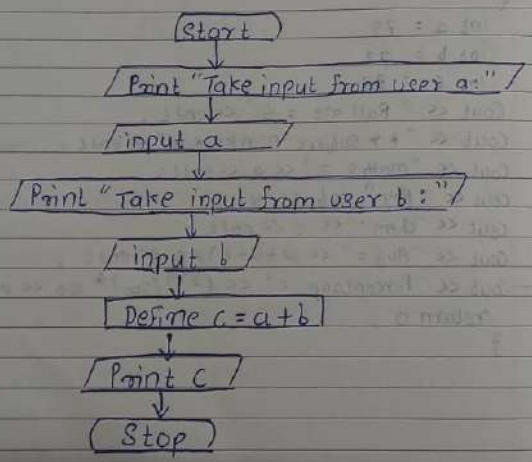
8] ~~Add 2 no's~~ (user input type)

Flow chart :-

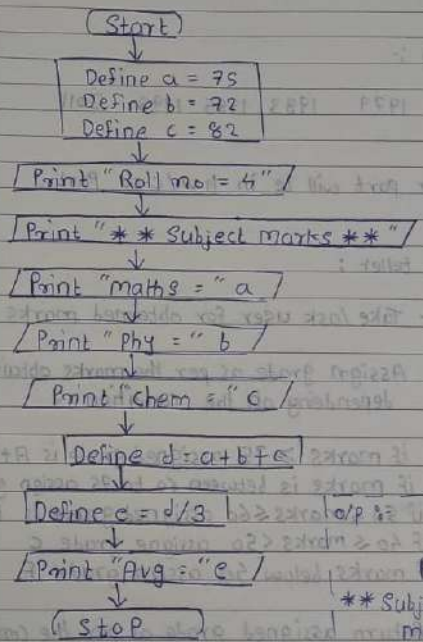


~~Start~~

C.W 8] Add 2 no's (user input type) :-



H.W :- 7] Soln in flow chart :-



o/p :-

Roll no = 4
** Subject marks **
maths = 75
Phy = 72
chem = 82
Avg = 76.3

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CW :- O/P :- Name : Anushree P
Class : FYIT
Roll no : 18
Grade : A

Points :- • else has no condition
• Braces optional for single statement
• Exactly one of two blocks run.

Algorithm :- Step 1 → Take variable m
Step 2 → Ask user his marks as input value for variable m
Step 3 → check if m > 60
Step 4 → if the if condition is true print assigned Grade as A if false print assigned Grade as F

Program :-
```
include <iostream>
using namespace std;
int main ()
{
Cout << "Enter your marks" <<endl;
~~Cout~~
Cin >> M

if(M >60)
 {
  Cout << "Grade = A" << endL;
 }

else
 {
  Cout << "Grade = F" << endL;
 }
```

Program :-
```
include <iostream>
using namespace std;
int main () {

 int marks = 70

 Cout << " Name : Anushree P" <<endL;
 Cout << "class : FYIT" <<endL;
 Cout << "Roll no : 18" << endL;
 Cout << "Grade : ";

 if (marks>60)
 {
  Cout << " A" <<endL ;
 }

 else
 {
  Cout << "B" <<endL ;
 }

 return 0 ;
}
```

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Unit II :

If statement :- executes the block only when the condition is true.

```
if (condition)
{
// execute if condition is true
}
```

- condition must evaluate to true or false
- Braces {} define block scope
- Block is skipped if condition is false.

eg:-

```
include <iostream>
using namespace std;
int main()
{
  int age = 20
  if (age >= 18)
   {
    Cout << " Eligible to vote";
   }
  return 0;
}
```

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Qu :- check if no is even or odd

Program :-

```
# include <iostream>
using namespace std;
int main () {

  int number ;
Cout << " Enter number ";
Cin >> number;

if (number %2 == 0)
{
 Cout << "number is even"<<endl;
}

else
{
 Cout << " number is odd "<<endl;
}

return 0;
}
```

Ladder else if :-

```
int marks = 72

if (marks >= 90)
{ cb
  Cout << "Grade A";
}
else if (mark >= 75)
 {
  Cout << "Grade B";
 }
else if (marks >= 60)
 {
   Cout << "Grade c" ;
 }
else
  {
    Cout << " Grade D" ;
  }
```

{I have not followed proper format in above program}

key points :-

* Conditions checked in order : first match wins
* Remaining conditions are skipped once matched
* Final else acts as a catch all default

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* for loop :- for iterations. ; first execution the check condition

for (i = n ; i <= n ; i--)

initial letter (start) — range/ condition — increment or decrement

Q factorial of number

```
#include <iostream>
using namespace std;

int main() {

int n
int fact = 1;
cout << "Enter the number to find factorial : ";
cin >> n;

for (i = n ; i >= 1 ; i-- )
{
  fact = fact * i;
}
cout << "factorial :" << fact;
return 0;
}
```

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Q user gives no. ; print the table

```
#include <iostream>
using namespace std;

int main() {

int n;
int mult = 1
cout << "Enter number n";
cin >> n;
for (i = mult ; i <= 10 ; i++)
{
   mult = mult + i * n
}
cout << "Table = " <<

cout << n << " x " << i << " = " << mult

}
```

Q user gives no. ; print the table

```
#include <iostream>
using namespace std;

int main() {

int n;

cout << "Enter number:";
cin >> n;

for (i = 1 ; i <= 10 ; i++)
{
   cout << n << " x " << i << " = " << i * n;
}
return 0;
}
```

note:- cout andar likha :. o/p need to be }

agar eek final value chaiye hogi to cout } ke bahar hoga.

```
2 x 1 = 2
2 x 2 = 4
: x : = :
```

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Nested if :-

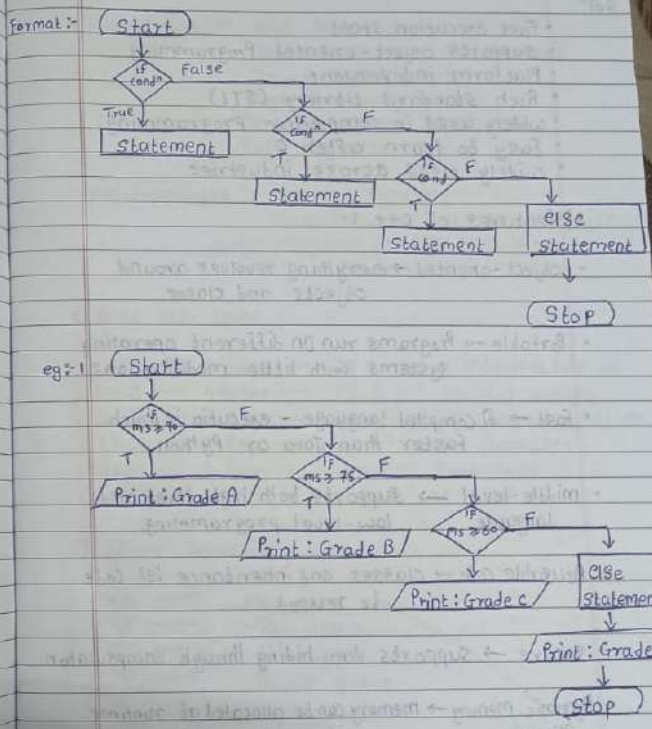
```
int age = 20;
char license = 'y';

if (age >= 18)
{
   if (license == 'y')
      cout << "Can drive";
   else
      cout << "Needs a license";
}
else
   cout << "Too young to drive";
```

Key points :-

- Inner if only evaluated when outer condition is true
- useful when a decision depends on two or more factors
- Deep nesting can hurt readability
  - Keep it shallow

Nested flowchart :-



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Q Sum of even no from 1 to no. n →

```
# include <iostream>
using namespace std ;

int main() {

int n ;
int ini ;
cout << "Enter no " ;
cin >> n ;

for (i = 2 , i <= n , i++)
{
  ini = i + 2
}
cout << " Sum of even no. " << " = " << ini ;
return 0 ;
}
```

Q " "

```
# include <iostream>
using namespace std ;

int main() {

int Sum = 0 ;
```

While loop :- entry controlled loop (check cond° before execution)

Syntax :- initialization

```
while (condition)
{
  # Statement
}
```

while loop :- looping statement in c++ used to execute a block of code statement repeatedly

working :- 1] condition is checked
2] if cond° is true, statement inside loops are executed.
3] cond° checked again
4] this continues until the condition becomes false
5] when the condition became false, the loop terminates.

agega aeo a sakte hai.

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Switch -> decision making statement :- similar to if/else

useful when comparing one variable with several fixed values.

Syntax :- switch (expression)
{
Case

Case

Case

case se match nai hua to { break;
Default
Default value ka cout jayega. }

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eg:- int main() {

```
int i = 1;

while (i <= 5)
{
  cout << i << endl;
  i++;
}

return 0;
}
```

" Convert for loops example into while loop.

Q print even no. from 1 to 20 using while loop

```
int main () {

 int i = 2

while (i <= 20)
{
  cout << i  << endl;
  i += 2;
}

return 0;
}
```

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